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11

How to Calculate the SWL of Ropes and Chains

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Wire rope and chain are the important part of the hoist which are closely bound up with the safe work load, now let's talk about how to calculate the SWL of ropes and chains.

As we all known:

All diameters are measured in millimetres (mm).

All weight are calculated in Kilograms(kg).

Formula SWL-Fibre Rope

Diameter in mm squared=The weight in kg.

$(Diameter)^2$=SWL Kg

Example: For a 12mm diameter rope.....

SWL=12*12=144Kg

Formula Size-Fibre Rope

Choosing the correct rope for the load

Firbre Rope Square root of weight

Example:900Kg load

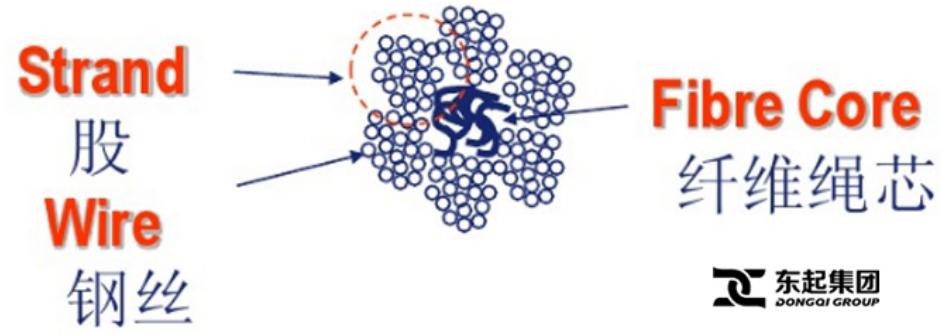
Rope diameter=√90=30mm

Wire Rope Construction

6*19 (6 strands*19 wires)

6*24 (6 strands*24 wires)

6*37 (6 strands*37 wires)



Formula SWL-Steel wire rope

SWL=$8D^2$

8 times stronger than Fibre

$(Diameter)^2$*8=SWL Kg

Example: For a steel wire rope of 15mm diameter

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SWL=15*15*8=1800Kg

Fomula Size-Steel wire rope

Square root of the load divide by 8

Example:5000kg load

Ø=√5000÷8

=√625

The diameter of steel wire rope=25mm

SWL=8D²

Steel wire rope descarding factor

10% of the wires broken in a length of eight times the diameter

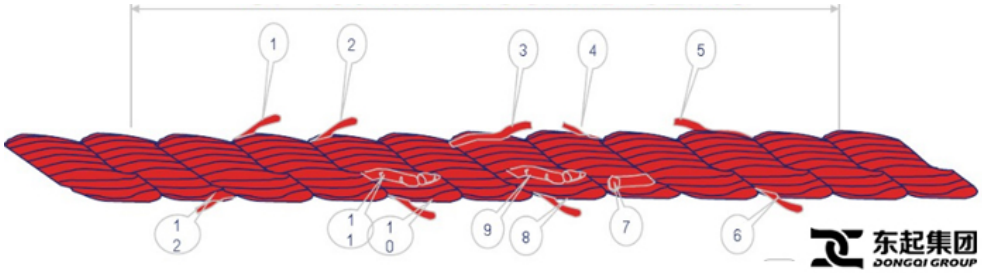
Example:6*19 construction 20mm Diameter

6*19=114 wires

10% of 114 wires=11

8*20mm=160mm

11 wires broken over a length of 160mm discard sling



Formula SWL-Chain grade 30

Nine times stronger than Fibre

(Diameter)²*9=SWL (Kg)

Contoh: Garispusat=18mm

SWL=18*18*9

=2916Kg

SWL=9D²

Formula SWL-Rantai Gred Peblagai

Grade 30 chain=9 times stronger than fibre rope

Grade 40 chain=12 times stronger than fibre rope

Grade 60 chain=18 times stronger than fibre rope

Grade 80 chain=24 times stronger than fibre rope

Formula for different grades

0.3*diameter²MM*Grade of Chain

Or SWL=0.3*D²*Grade

Formula Size-Chain of different grade

Square root of the wight(Kg) divide by the grade of China

Example:Weight=5400Kg & Grade=40

Diameter of Chain=√Weight÷(0.3*Grade)

=√5400÷(0.3*40)

=21.2 mm

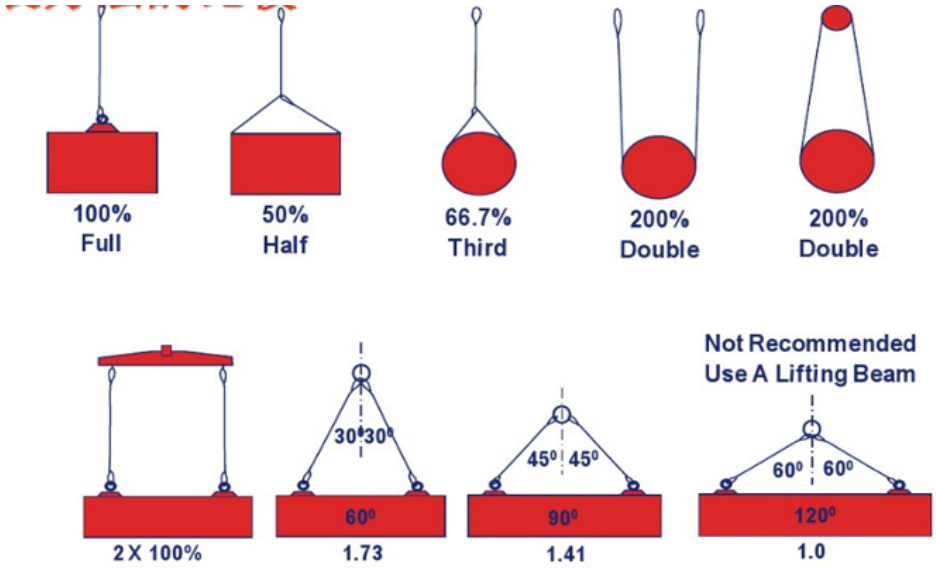
W=12D²

Load factor 2 slings at an angle

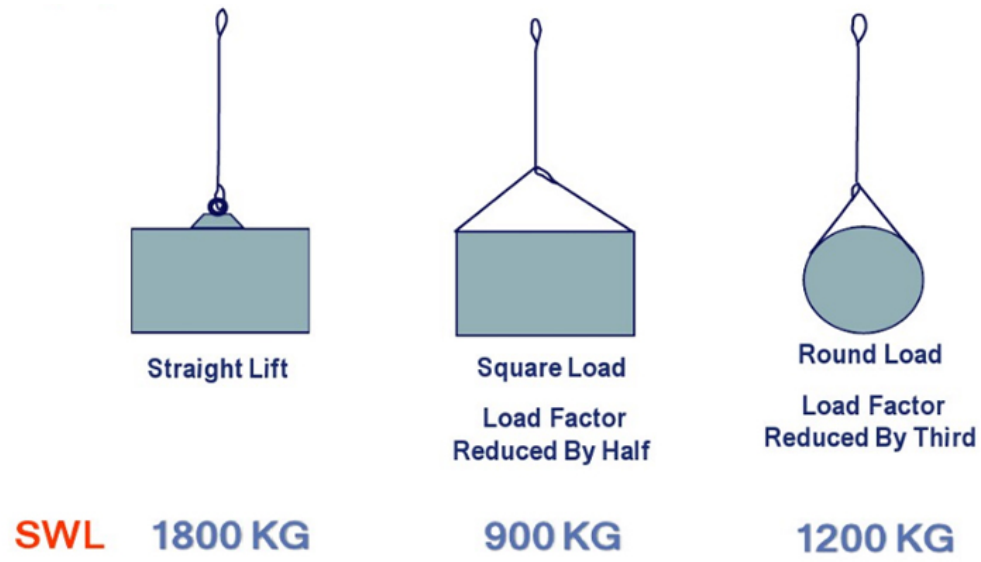
Sling angle 30° 60° 90°

Load factor 1.93 1.73 1.41

Comparative Methods of Slinging



Full Wrap



So, now do you have more understanding about it? If you are still not clear with that but want to know more welcome to contact us and ask more information.

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Dongqi Group is a Sino-foreign joint venture. The Group consists of subsidiaries: Henan Dongqi Machinery Co., Ltd. (manufacturing), Henan Yuantai Crane Machinery Import&Export Co., Ltd. (Foreign Trade).

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Factory establish in -- 1984. With 33 years fabrication experience, we are professional.

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